Training and Evaluation — Results

Machine Learning Driven Assessment of Global Life Expectancy Trends
WHO Life Expectancy Dataset (193 countries, 2000–2015)

Experiment Tracking (Weights & Biases)

All training runs for this project were logged to the Weights & Biases platform. The project is public and available at:

https://wandb.ai/ml-group-j/life-expectancy-ml

The project contains four logged runs, one per model developed by the group: *linear-regression*, *lasso*, *random-forest*, and *xgboost*. Each run records the model's hyperparameter configuration (e.g., the cross-validated LASSO alpha, and the best parameters found by GridSearchCV for the ensemble models) together with its final test-set metrics (R^2, MAE, RMSE). For the XGBoost model, the run additionally includes full training curves — train and test RMSE logged over all 300 boosting rounds.

Model Comparison — Test-Set Results

Model	R^2	MAE (years)	RMSE (years)
Linear Regression (baseline)	0.82	3.0	4.1
LASSO (cross-validated α)	0.8195	2.976	4.108
Random Forest (GridSearchCV)	0.9679	1.050	1.667
XGBoost (GridSearchCV)	0.9718	0.954	1.562

Table 1: Test-set performance of the four models. XGBoost achieves the best result on all three metrics (highlighted). The tree-based ensembles roughly halve the prediction error of the linear baselines, reducing RMSE from ≈4.1 to ≈1.6 years.

The linear baseline already explains ≈82% of the variance in life expectancy, indicating that a large share of the signal in the WHO indicators is linear and additive. The additional ≈15 percentage points of R^2 gained by Random Forest and XGBoost quantify the contribution of non-linear effects (e.g., HIV/AIDS prevalence, adult mortality) and feature interactions that linear models cannot capture. These results are consistent with published Kaggle baselines on this dataset, which typically report $R^2 \approx 0.80$–0.85 for linear models and $R^2 \approx 0.94$–0.96 for tuned tree ensembles.

Group Model Comparison (W&B Charts)

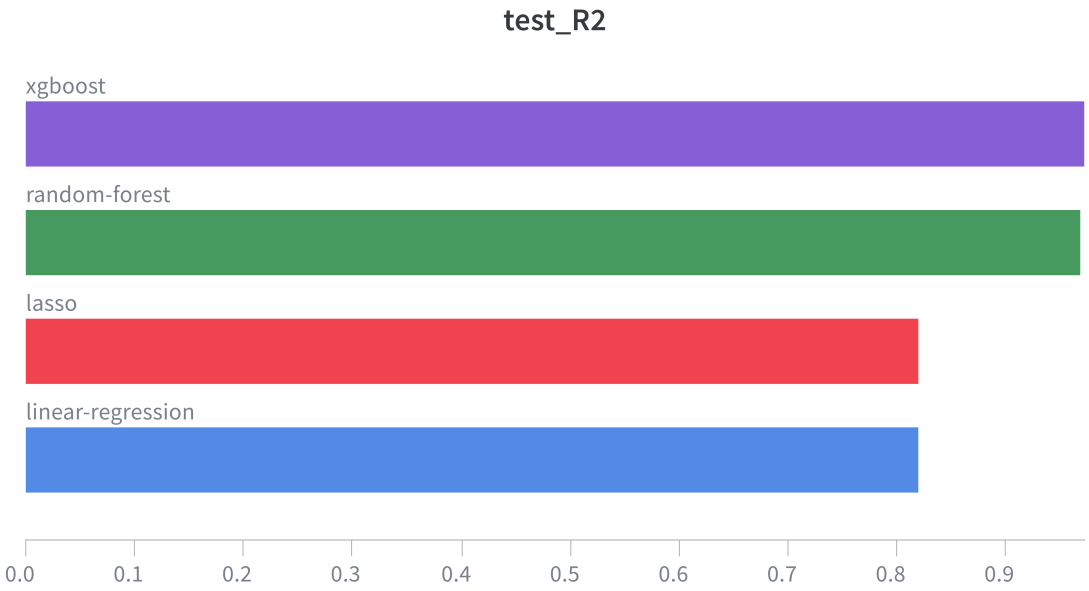


Figure 1a: Test-set R^2 by model (higher is better).

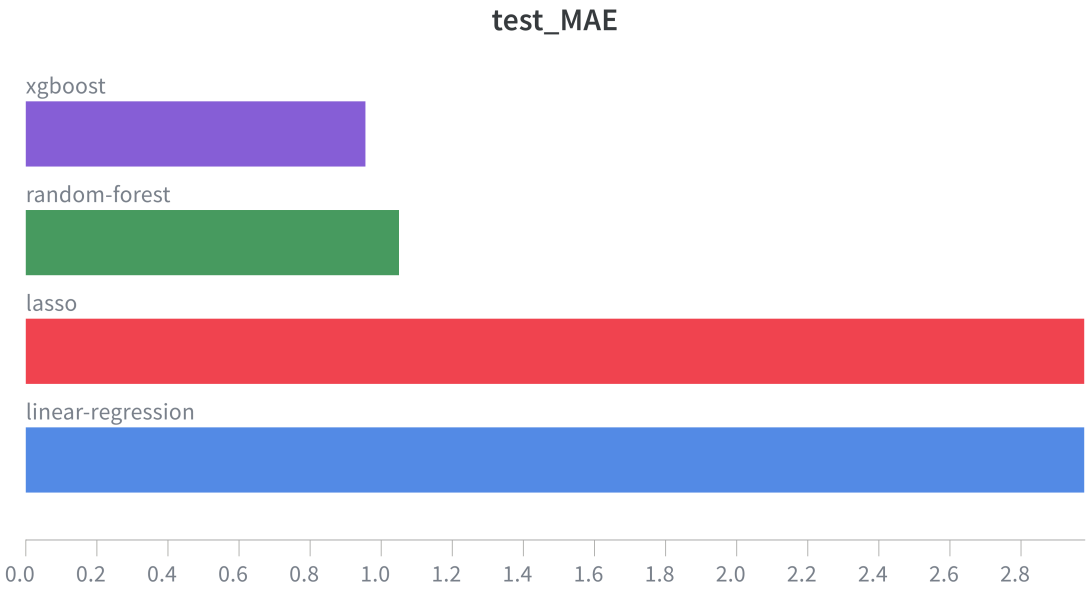


Figure 1b: Test-set MAE in years by model (lower is better).

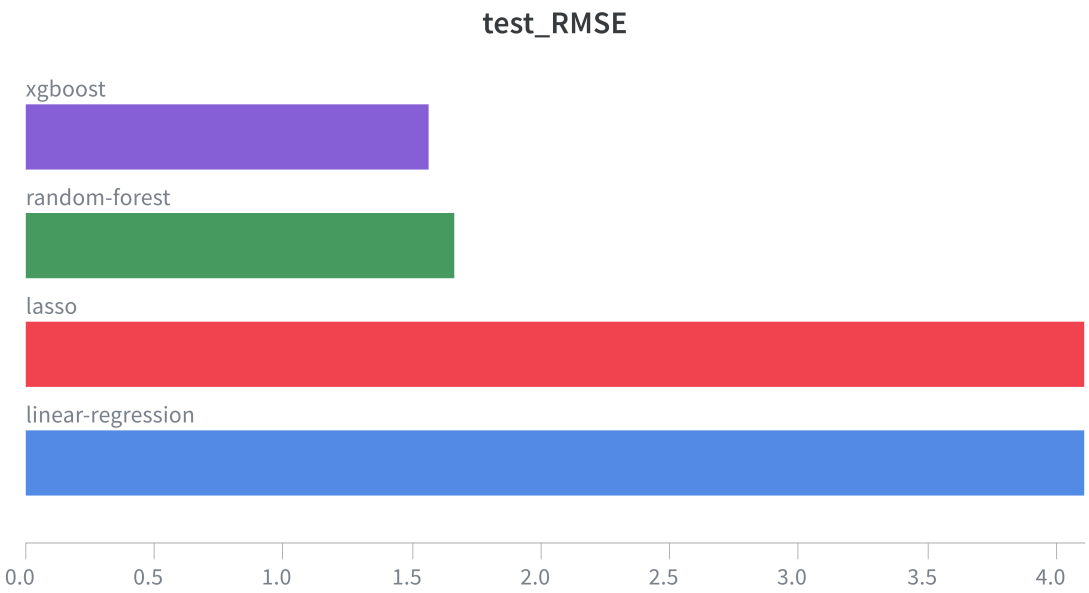


Figure 1c: Test-set RMSE in years by model (lower is better). Screenshots from our public W&B project.

XGBoost Training Logs

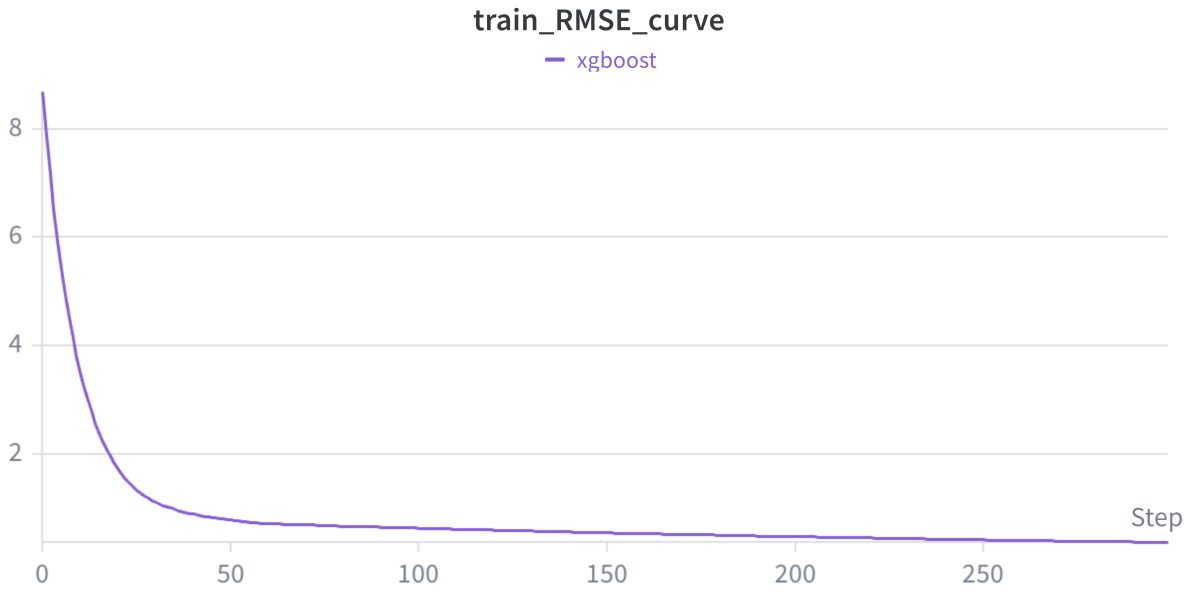


Figure 2a: XGBoost training RMSE over 300 boosting rounds (converges to ≈0.37 years).

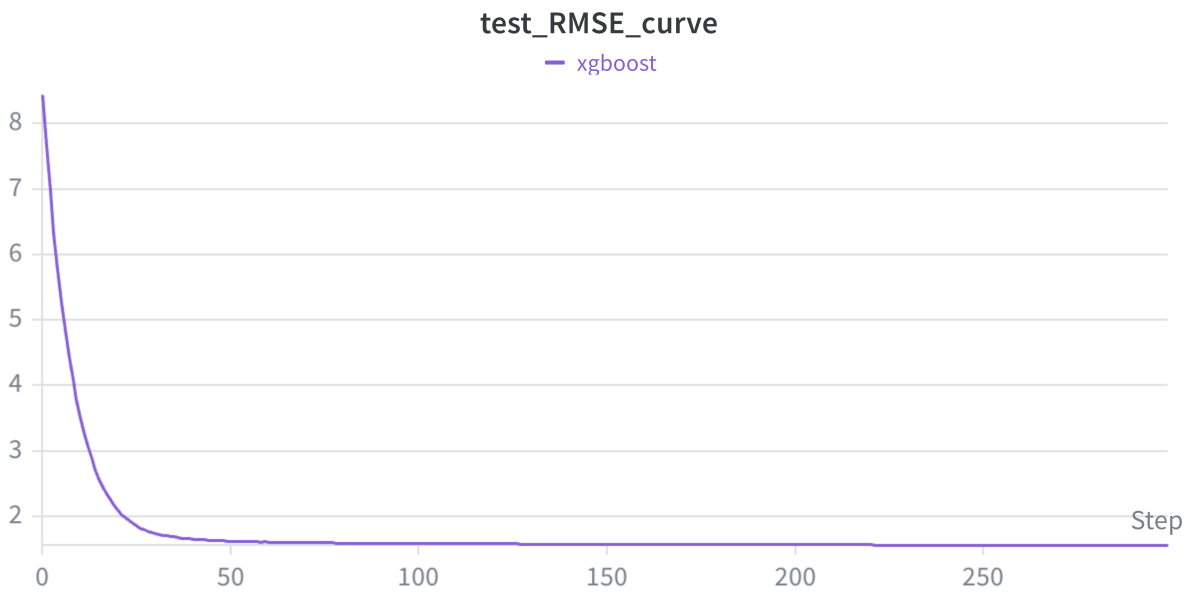


Figure 2b: XGBoost test RMSE over 300 boosting rounds (plateaus at ≈1.56 years).

The training curves show rapid convergence within the first ≈50 boosting rounds, after which test RMSE plateaus while training RMSE continues to decrease slowly. The resulting gap between final training error (≈0.37 years) and test error (≈1.56 years) indicates a degree of overfitting despite 5-fold cross-validated hyperparameter selection. A further caveat is that the random row-wise train/test split allows the same country to appear in both sets across different years, so performance on entirely unseen countries would likely be somewhat lower.